

Agentic AI for Materials Discovery

Chanyoung Park

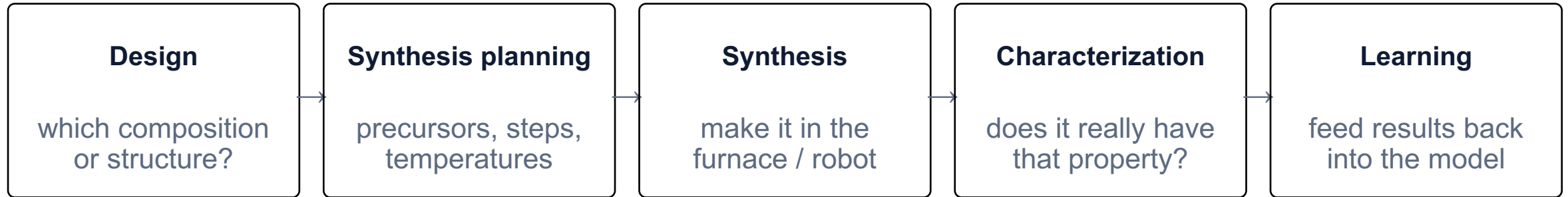
Associate Professor, KAIST

Industrial and Systems Engineering
Graduate School of Data Science
Graduate School of AI

cy.park@kaist.ac.kr

The materials discovery pipeline

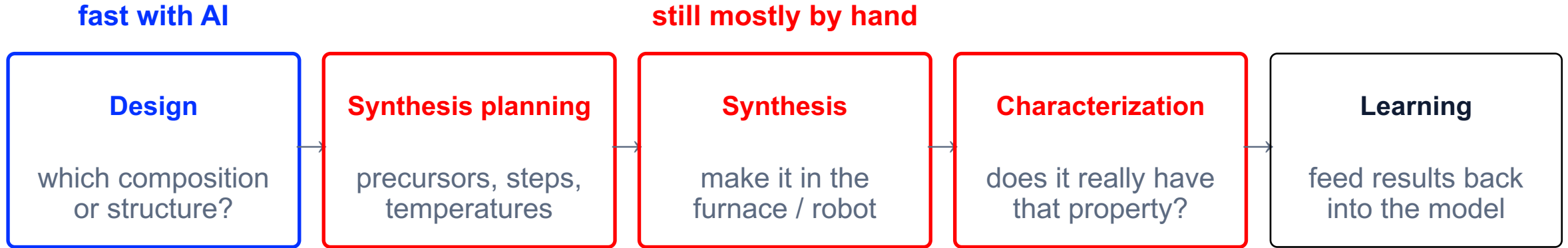
five stages, from a target composition to a verified material



- One pass takes days to weeks, and most of that time is spent in synthesis and characterization
- Each pass also produces data: which recipe worked, which did not, and what was measured
- AI can support every stage, but today the five stages are at very different levels of maturity

Where AI stands on this pipeline today

fast at design, still manual after it

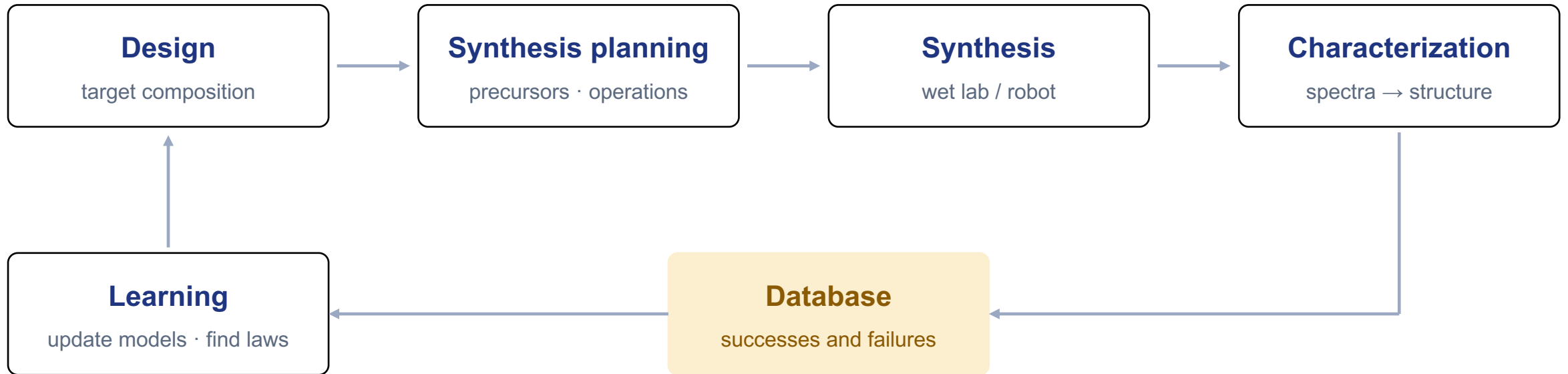


The bottleneck is validation, not ideas

- Generative models propose millions of candidates a day; synthesis and characterization are still by hand
- Candidates pile up unverified: most 'N new materials' in papers are predictions, not real measurements

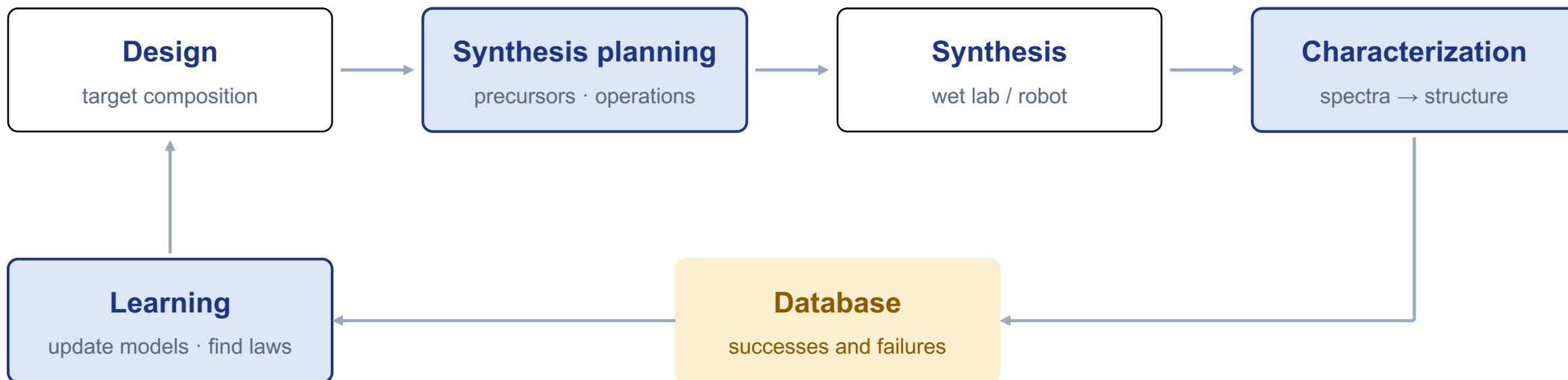
The materials discovery loop

- In practice it is a **loop**: every result flows back as training data



Where our work fits in the loop

- Today: **three stages** of the loop where our lab has built AI systems



Design and synthesis are outside today's scope — our contributions are in the highlighted stages

Synthesis planning

- Retrieval-Retro (NeurIPS 2024)
- MSP-LLM (NeurIPS 2026)
- Retro-Forge (ICML 2026 AI4Sci WS)

Characterization

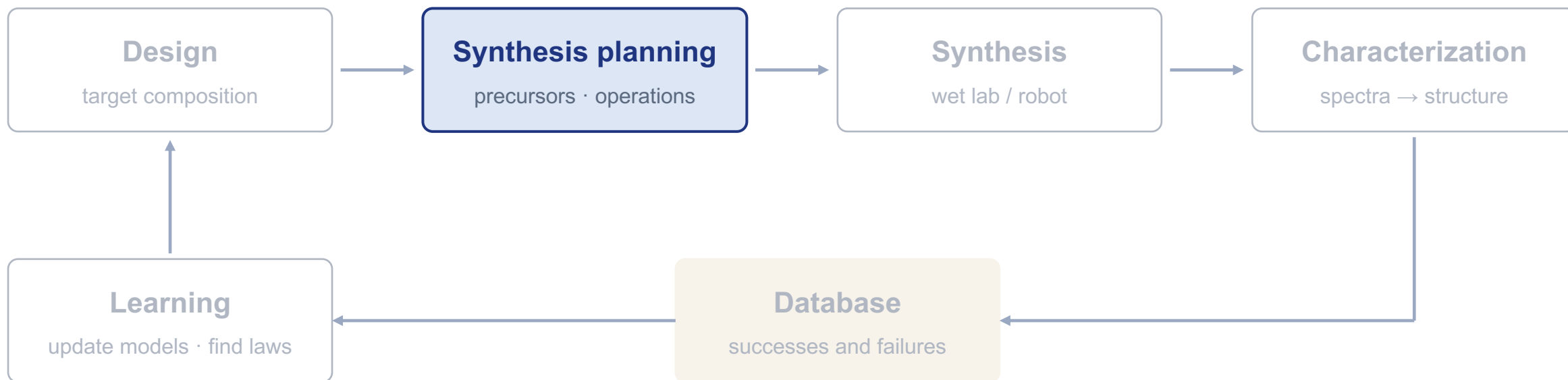
- IR-Agent (ICLR 2026)
- PCSP (NeurIPS 2026)

Learning

- Multi-agent LLM equation discovery (under review)

Part 1 — Synthesis planning

- Part 1: **turning a target into a recipe** — Retrieval-Retro, MSP-LLM, Retro-Forge



Given a target composition, what do we mix, and how do we heat it?

Synthesis planning

- Retrieval-Retro (NeurIPS 2024)
- MSP-LLM (NeurIPS 2026)
- Retro-Forge (ICML 2026 AI4Sci WS)

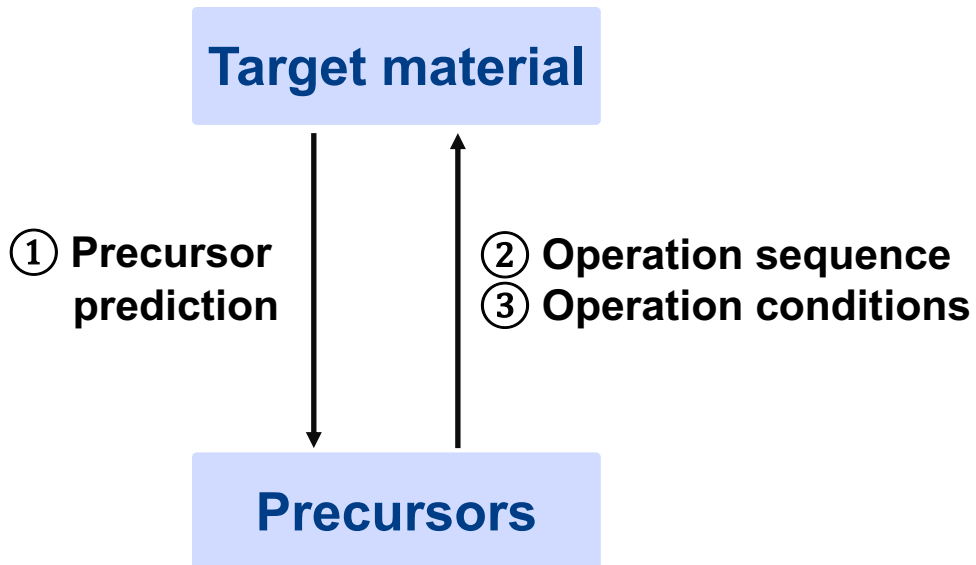
Characterization

- IR-Agent (ICLR 2026)
- PCSP (NeurIPS 2026)

Learning

- Multi-agent LLM equation discovery (under review)

Synthesis planning: three sub-problems



① Precursor prediction

→ Retrieval-Retro (NeurIPS 2024)

→ Retro-Forge (ICML 2026 AI4Sci WS)

① + ② Precursors + Operation sequence prediction

→ MSP-LLM (NeurIPS 2026)

① + ② + ③ Precursors + Operation sequence + Conditions

→ ongoing work

Example

- Target material: $\text{Mg}_{0.5}\text{Zn}_{0.5}\text{Fe}_2\text{O}_4$ (Mg-Zn ferrite)
- ① precursors: Fe_2O_3 , MgCO_3 , ZnO
- ② sequence: dry → mix → heat → mix → shape → heat → anneal...
- ③ conditions: temperature / time / pressure at each step

Step	Temp.	Time	Key point
Drying	100–120°C	1–2 h	Remove water
Heat 1	750–850°C	2 h	MgCO_3 decomp.
Heat 2	950–1050°C	3–5 h	Spinel forms
Anneal	800–900°C	1–3 h	Stabilize
Press	100–300 MPa	—	Densify

Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge

NeurIPS 2024

Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Chanyoung Park

Retrieval-Retro: why copying recipes is not enough

(a) Subset case

Target Material: $\text{Ca}_3\text{Bi}(\text{PO}_4)_3 \rightarrow \text{PH}_6\text{NO}_4, \text{Bi}_2\text{O}_3, \text{CaCO}_3$
 Reference Material: $\text{BiPO}_4 \rightarrow \text{PH}_6\text{NO}_4, \text{Bi}_2\text{O}_3$

Shared Precursors

(b) New case

Target Material: $\text{Li}_3\text{Mg}_7 \rightarrow \text{MgH}_2, \text{LiH}, \text{Mg}$
 Reference Material: $\text{Li}_4\text{MgV}_5\text{O}_{10} \rightarrow \text{MgO}, \text{Li}_2\text{CO}_3, \text{V}_2\text{O}_3$

New Precursors

(d) Thermodynamic Forces

Precursor Sets	ΔG
$\text{AO}, \text{AO}_2, \text{BCO}$	-400 meV
$\text{A}_2\text{CO}_2, \text{BO}_2$	-100 meV

Target Material: $\text{A}_2\text{BCO}_4 \rightarrow$

(c) Statistics

90.3%

9.7%

Subset

New

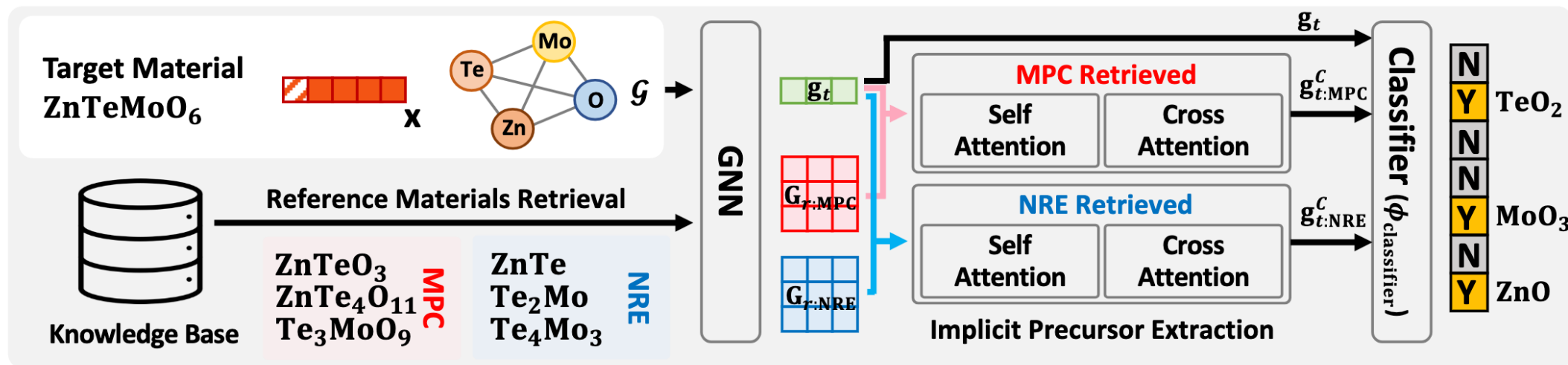
Domain knowledge

- **The obvious approach: retrieve and copy**
 - Chemists borrow precursors from a similar material
 - Retrieval: nearest reference $\rightarrow$ copy its precursor set
- **Problem 1: new routes cannot be copied**
 - (a) shares some precursors: copying works
 - (b) shares none: $\sim 10\%$ of the data, the new routes
- **Problem 2: thermodynamics is ignored**
 - (d) precursor sets for the same target differ in ΔG^*
 - More negative $\Delta G \rightarrow$ the reaction is more likely
 - Prior work retrieves by similarity only

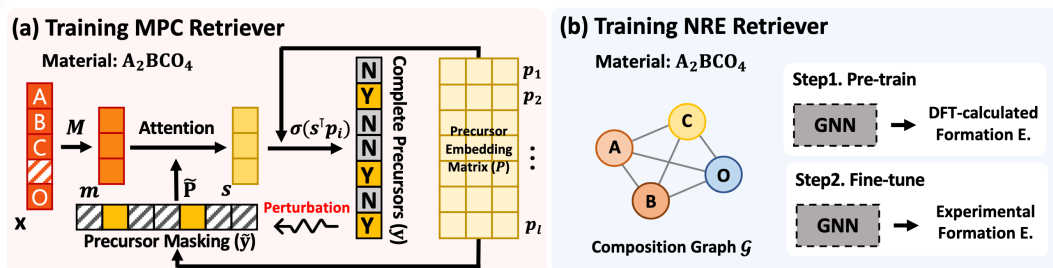
* ΔG : Gibbs free energy of the reaction; $\Delta G < 0$ means the reaction can proceed

Goal of Retrieval-Retro: bring in thermodynamics, and propose recipes that nobody has written down before

Retrieval-Retro: physics-guided implicit retrieval



1. Which reference materials to retrieve?



"materials likely to use a similar precursor combination"

reaction feasibility (ΔG) as the retrieval criterion

→ Retrieval guided by physics, not only by similarity

2. Implicit extraction, never copying

- Reference materials are encoded with a GNN, and the target attends over them
- Precursor information is extracted in the representation space, not copied

→ Combinations that no single reference contains

Retrieval-Retro: largest gains on the year split

Model	Interaction Retrieval		(a) Year Split						(b) Random Split					
			Top-K Accuracy				Recall		Top-K Accuracy				Recall	
			Top-1	Top-3	Top-5	Top-10	Macro	Micro	Top-1	Top-3	Top-5	Top-10	Macro	Micro
Composition MLP	✗	✗	31.60 (1.70)	34.37 (1.58)	35.22 (1.43)	36.56 (0.160)	31.42 (0.030)	31.44 (0.060)	58.56 (0.47)	62.20 (0.36)	62.95 (0.029)	64.32 (0.42)	54.56 (0.44)	55.35 (0.57)
He et al. [12]	✗	✓	45.03 (1.85)	48.02 (1.86)	49.11 (1.77)	51.09 (1.93)	44.72 (1.83)	44.75 (1.86)	61.94 (1.5)	66.44 (1.48)	67.46 (1.55)	68.84 (1.65)	58.55 (1.45)	59.35 (1.34)
ElemwiseRetro	✓	✗	53.45 (0.58)	57.07 (0.52)	58.19 (0.72)	60.84 (0.78)	53.12 (0.60)	53.19 (0.60)	77.23 (0.70)	80.93 (0.54)	81.57 (0.67)	82.78 (0.64)	72.33 (1.14)	73.26 (0.99)
Roost	✓	✗	54.38 (0.75)	57.82 (0.81)	58.82 (1.00)	60.71 (1.15)	54.01 (0.75)	54.04 (0.74)	78.42 (0.91)	82.32 (0.91)	83.07 (0.83)	84.10 (0.66)	73.38 (1.41)	74.46 (1.22)
CrabNet	✓	✗	57.15 (0.77)	61.60 (0.85)	62.44 (0.82)	64.14 (0.86)	56.79 (0.77)	56.82 (0.77)	78.69 (0.78)	81.62 (0.74)	82.27 (0.67)	83.35 (0.56)	73.27 (1.21)	74.28 (0.99)
Graph Network	✓	✗	58.95 (0.41)	63.10 (0.63)	64.07 (0.68)	66.30 (0.62)	58.54 (0.42)	58.61 (0.41)	77.91 (1.31)	81.55 (0.98)	82.37 (0.92)	83.50 (0.90)	72.96 (1.53)	73.88 (1.29)
Graph Network + MPC	✓	✓	<u>60.01</u> (1.10)	<u>64.15</u> (1.10)	<u>65.15</u> (1.17)	<u>67.19</u> (0.83)	<u>59.61</u> (1.10)	<u>59.66</u> (1.10)	<u>79.09</u> (1.25)	<u>82.95</u> (1.13)	<u>83.82</u> (1.19)	<u>84.97</u> (0.94)	<u>73.86</u> (1.34)	<u>74.81</u> (1.23)
Retrieval-Retro	✓	✓	61.16 (0.38)	65.92 (0.71)	67.18 (0.76)	69.45 (1.03)	60.97 (0.62)	61.06 (0.62)	79.81 (0.68)	83.62 (0.77)	84.46 (0.78)	85.70 (0.88)	74.61 (0.98)	75.49 (0.89)

- **The year split is harder but realistic:** performance is lower than in the random split, and it better reflects predicting recipes for future materials
- **Retrieval matters most on the year split:** models that use reference materials consistently outperform non-retrieval models
- **Retrieval-Retro gives the largest gain on the year split:** best Top-1/3/5/10 accuracy and macro/micro recall, with a clear improvement over Graph Network + MPC

A Structured LLM Framework for Inorganic Material Synthesis Planning

NeurIPS 2026

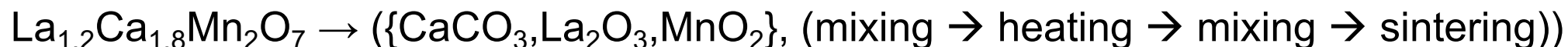
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, Chanyoung Park

MSP-LLM: one LLM for the complete synthesis plan

- Material Synthesis Planning (MSP)

$$f : \mathcal{M} \rightarrow 2^{\mathcal{M}} \times \mathcal{O}^*$$

target material $\rightarrow$ (precursor set, synthesis operation sequence)



- Prior work**: focuses on a single task — Precursor Prediction (PP) or Synthesis Operation Prediction (SOP)
- This work**: a unified LLM framework that solves the complete MSP end-to-end
- Why an LLM?
 - All inputs are text (chemical formulas) / LLMs carry broad physics/chemistry knowledge / reasoning capability
 - But, there is no MSP-specific fine-tuning strategy yet

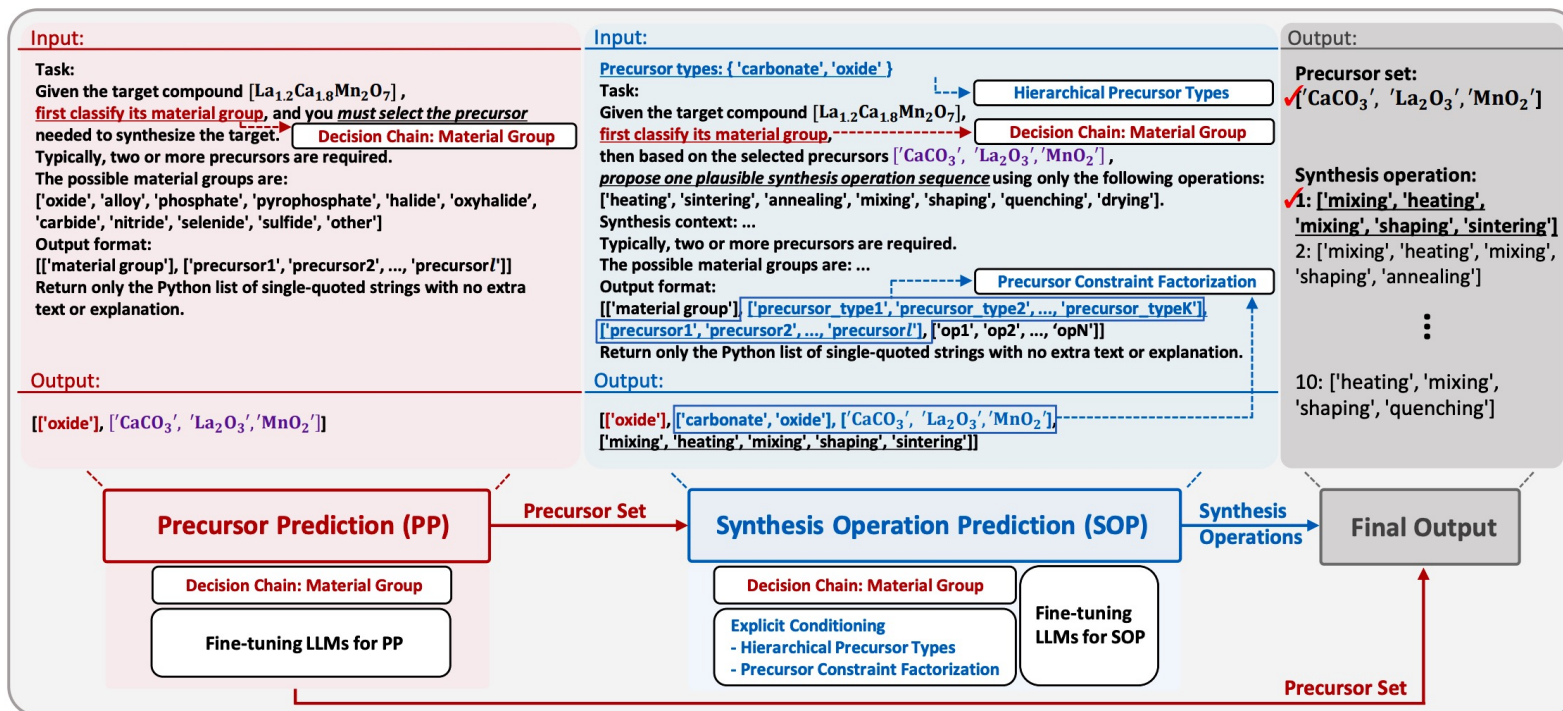
MSP-LLM: two stages, one decision chain

- Overall flow: target → precursors → operations, both stages with a fine-tuned LLM
 - Stage 1, Precursor Prediction: target → {CaCO₃, La₂O₃, MnO₂}
 - Stage 2, Synthesis Operation Prediction: target + precursors → [mixing, heating, ..., sintering]
- Two main ideas
 - Two stages: the operations are predicted conditioned on the precursors
 - Decision chain: predict the material group first, then the precursors conditioned on it → a chemically plausible subspace

MSP-LLM

To synthesize
La_{1.2}Ca_{1.8}Mn₂O₇,

what precursor set
and synthesis
operations
should I use?



MSP-LLM: explicit conditioning on the precursors

▪ Finding: the LLM ignores precursors in the prompt

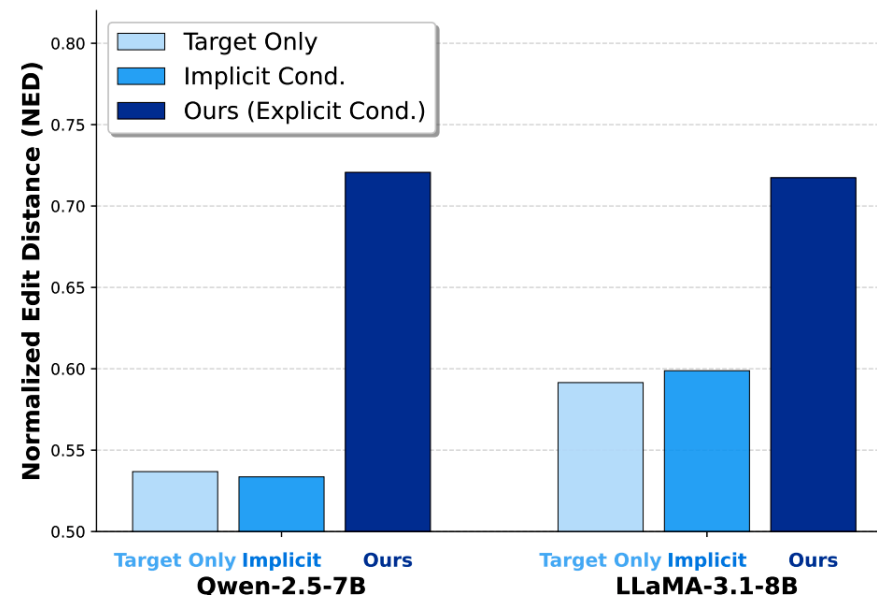
- Target-only: target material → operation sequence
- Implicit conditioning: precursor set in the input only → no gain

▪ Why?

- Precursor formulas are long, and several are listed in a row
- The LLM only has to predict the next token: if the loss can go down without the precursors, it ignores them

▪ Fix: Explicit conditioning

- Precursor set in the output as well as the input: the model must reproduce the precursors before the operations
- Plus precursor types (CaCO₃ → carbonate): a carbonate implies a heating step to drive off CO₂
- Learned once, and it transfers to every new carbonate



```
P = {CaCO3, La2O3, MnO2}
Types = {carbonate, oxide}
```

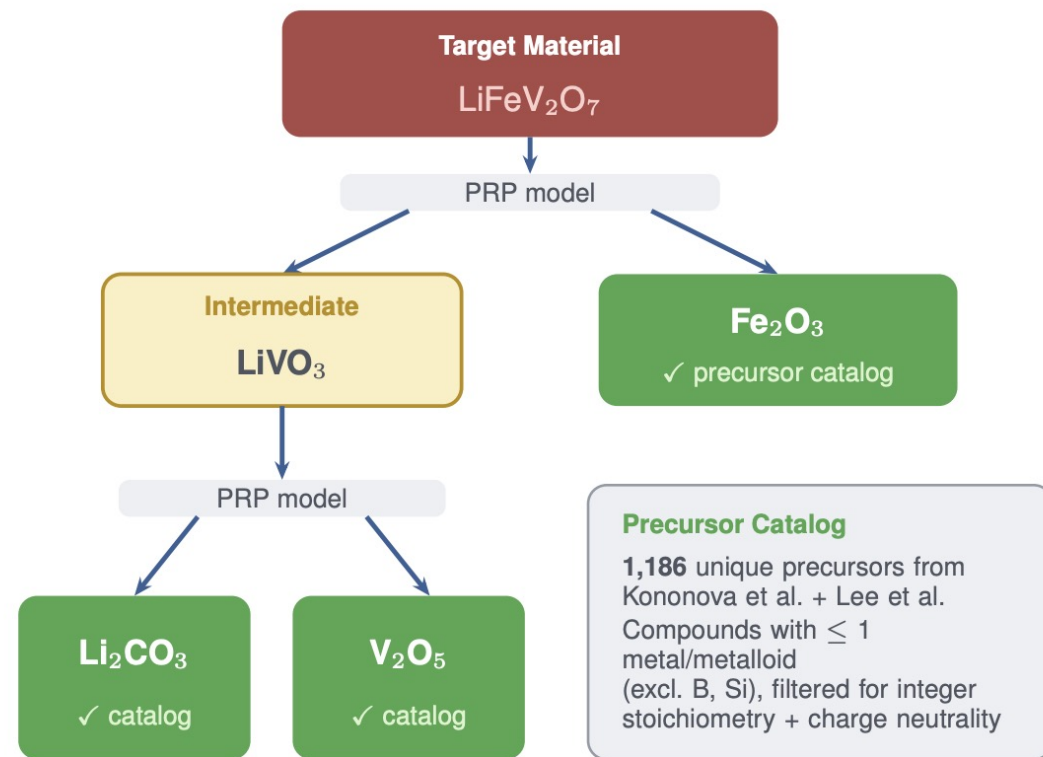
MSP-LLM: 18–23% Top-10 on the complete plan

Method	Split 1				Split 2			
	Top-1	Top-3	Top-5	Top-10	Top-1	Top-3	Top-5	Top-10
<i>Large Language Models (API)</i>								
GPT-4o-mini (20-shot)	0.0	0.09	0.09	0.19	0.19	0.28	0.28	0.47
GPT-4o-mini (40-shot)	0.0	0.0	0.09	0.09	0.27	0.27	0.37	0.64
GPT-4o (20-shot)	0.27	0.55	0.64	0.73	0.47	0.47	0.47	0.56
GPT-4o (40-shot)	0.27	0.27	0.37	0.64	0.37	0.84	0.93	1.12
GPT-5.1 (20-shot)	0.46	0.64	0.73	1.01	0.65	0.75	0.75	0.75
GPT-5.1 (40-shot)	0.55	0.64	0.64	0.73	0.37	0.37	0.47	0.47
Claude Haiku 4.5 (20-shot)	0.56	0.65	0.65	0.74	0.56	0.65	0.65	0.75
Claude Haiku 4.5 (40-shot)	0.56	0.65	0.65	0.74	0.37	0.84	0.93	1.12
GPT-5.1(PP) + GPT-4o-mini (SOP)	0.91	1.65	2.01	3.75	1.4	2.71	3.64	5.41
MSP-LLM (Qwen-2.5-7B)	5.76	10.79	13.62	18.56	<u>6.62</u>	13.34	16.79	<u>22.48</u>
MSP-LLM (LLaMA-3.1-8B)	<u>4.75</u>	<u>9.69</u>	<u>13.35</u>	<u>18.19</u>	6.72	<u>12.12</u>	<u>16.60</u>	23.23

- **Off-the-shelf API models almost always fail; MSP-LLM reaches 18–23% Top-10**
- **Still, the absolute performance is low**
 - A prediction counts as correct only if both the precursors and the operations are right
- **The two stages are chained: wrong precursors → the operation prediction fails too**
 - Improving precursor prediction is the natural next step, and the focus of our next work (Retro-forge)

Retro-Forge: beyond single-shot precursor prediction

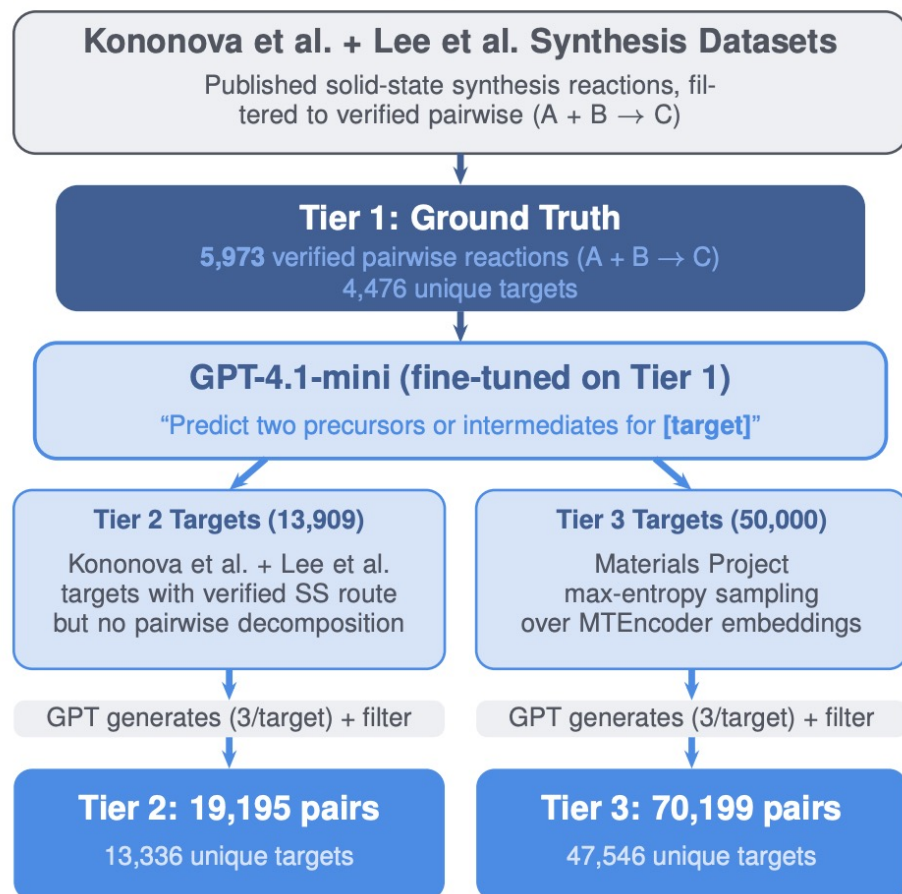
- **MSP-LLM predicts the precursor set in one shot**
 - $\text{LiFeV}_2\text{O}_7 \rightarrow \{\text{Li}_2\text{CO}_3, \text{V}_2\text{O}_5, \text{Fe}_2\text{O}_3\}$
- **But the actual route goes through an intermediate**
 - $\text{Li}_2\text{CO}_3 + \text{V}_2\text{O}_5 \rightarrow \text{LiVO}_3$, and then $\text{LiVO}_3 + \text{Fe}_2\text{O}_3 \rightarrow \text{LiFeV}_2\text{O}_7$
 - The intermediate decides the heating and annealing conditions
 - A route is more physically grounded than a bare precursor set
- **Real solid-state synthesis is pairwise and sequential**
 - Retro-Forge models it that way (right): one product $\rightarrow$ two reactants



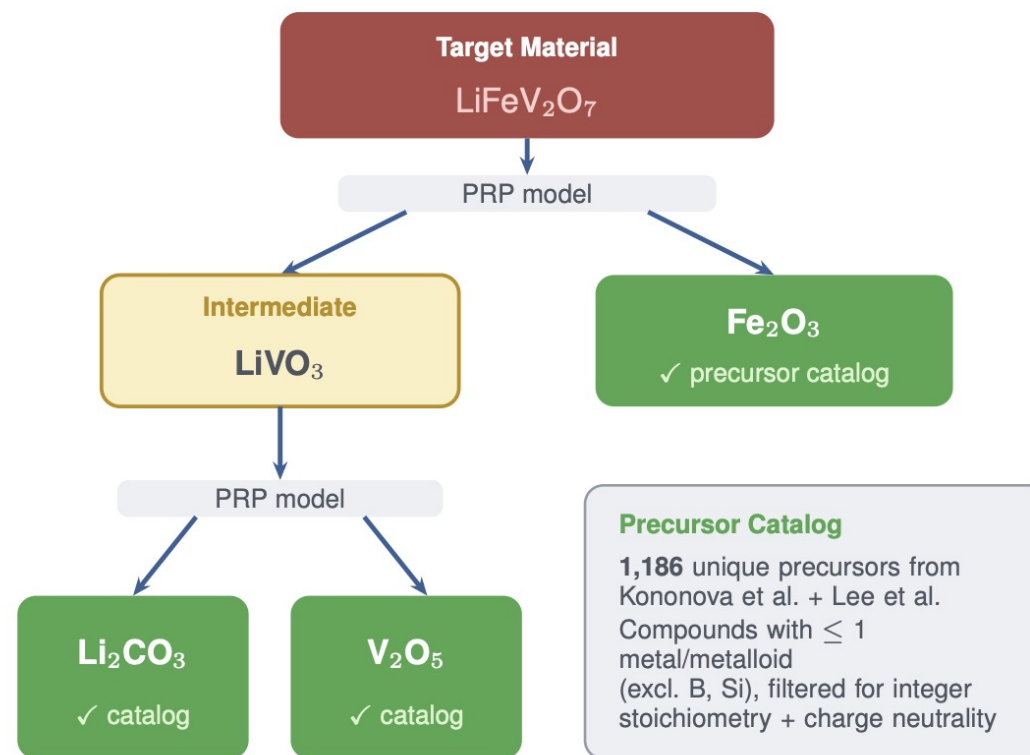
Retro-Forge: recursive pairwise retrosynthesis

- Core idea: Single-shot PP $\rightarrow$ Pairwise Route Discovery

(a) Data Augmentation Pipeline

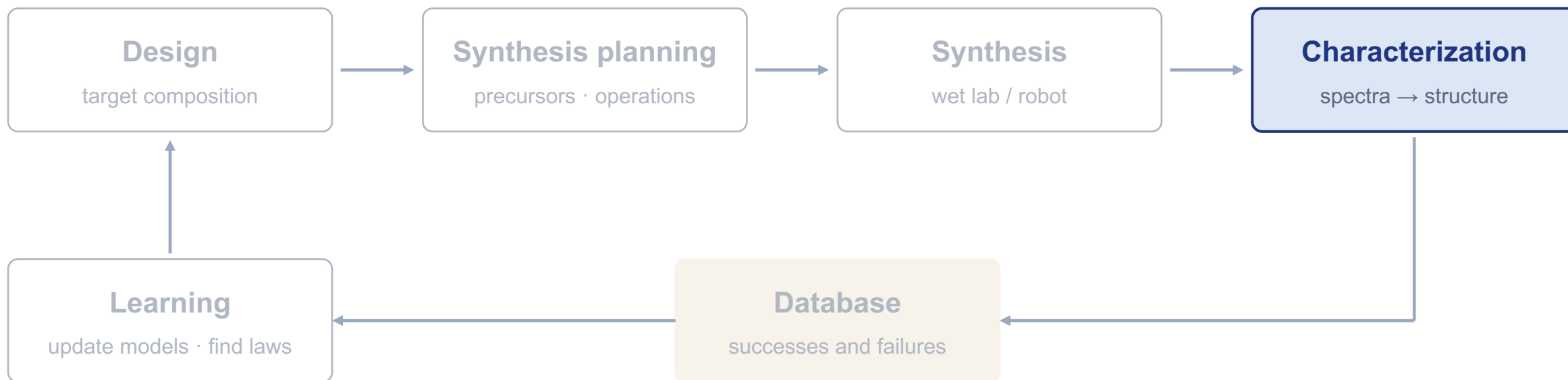


(b) Multi-Step Pairwise Retrosynthesis



Part 2 — Characterization

- Part 2: **reading what came out of the furnace** — IR-Agent



The instrument returns a spectrum, not a structure — how do we get the structure back?

Synthesis planning

- Retrieval-Retro (NeurIPS 2024)
- MSP-LLM (NeurIPS 2026)
- Retro-Forge (ICML 2026 AI4Sci WS)

Characterization

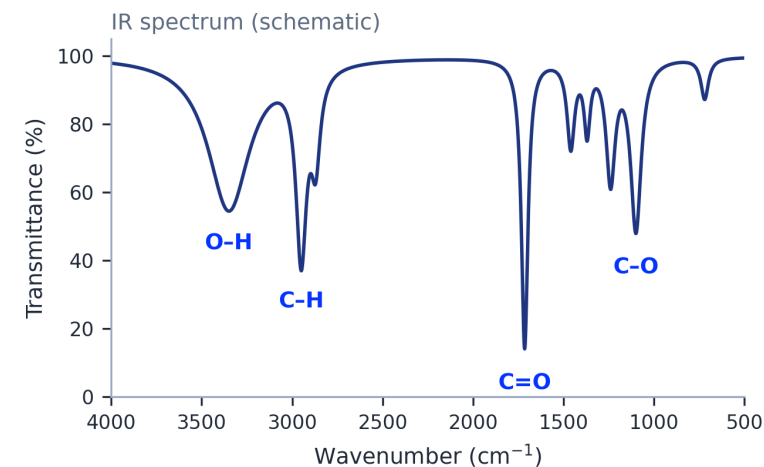
- IR-Agent (ICLR 2026)
- PCSP (NeurIPS 2026)

Learning

- Multi-agent LLM equation discovery (under review)

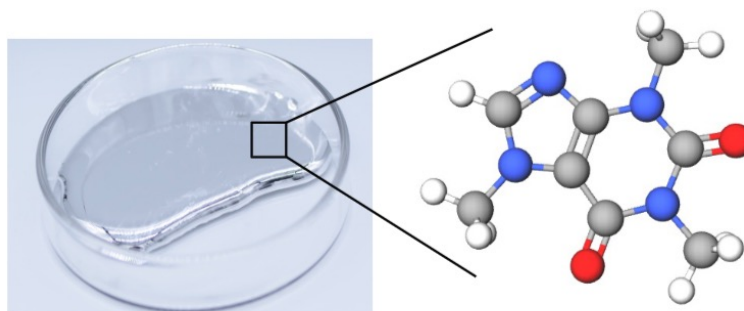
Spectroscopic data: an indirect view of structure

- **A spectrum: the signal from matter interacting with energy**
 - Shine light on a material, and record how it responds at each wavelength
 - Each bond vibrates at its own energy, so the peaks reveal the bonds present
- **Common types**
 - Infrared (IR), Raman, NMR, mass spectrometry (MS)

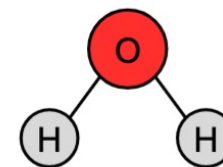


Infrared spectrometer

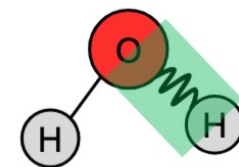
Infrared (IR)
 $\left(\approx \text{Energy}, E = \frac{hc}{\lambda} \right)$



target material



ground-state
atomic bonds



excited-state
atomic bonds

→ Spectroscopy is how we learn about a structure we cannot see directly

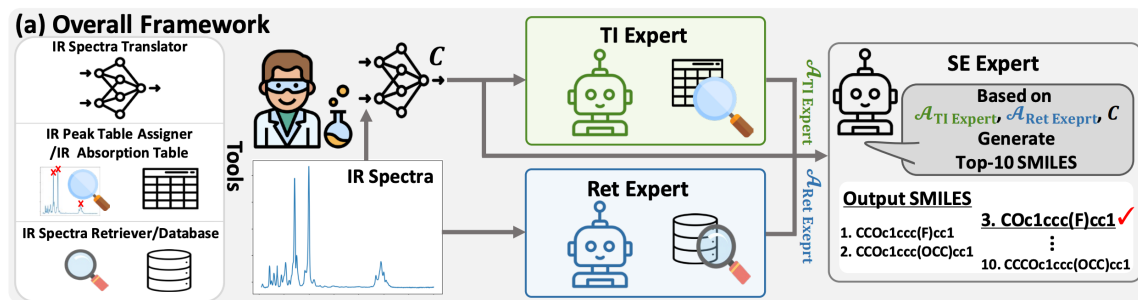
Why is structure inference from spectra hard?

▪ Three reasons it is hard

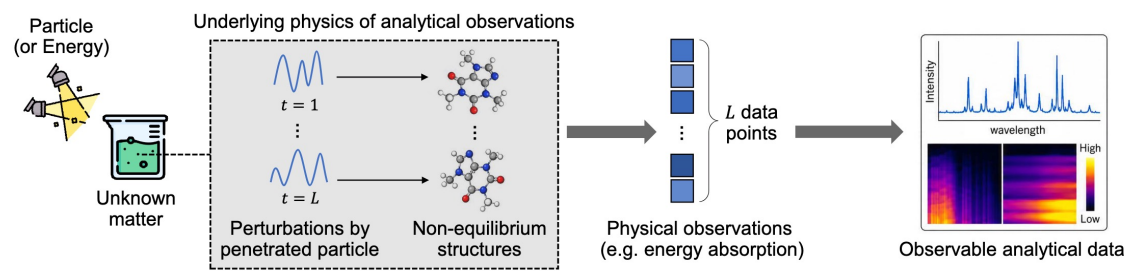
- Inverse problem: we observe a signal and have to reason backwards
- Ambiguity: different structures give similar spectra, and a single peak never decides anything
- Aggregation: a spectrum is a sum over many vibrational modes, plus the environment, plus noise

▪ So what inductive bias should the model have? Two routes

- 1. Model the way an expert interprets a spectrum → **IR-Agent (ICLR 2026)** (today)
- 2. Model the physical process that generates the spectrum → **PCSP (NeurIPS 2026)**



IR-Agent: expert-style reading of the spectrum



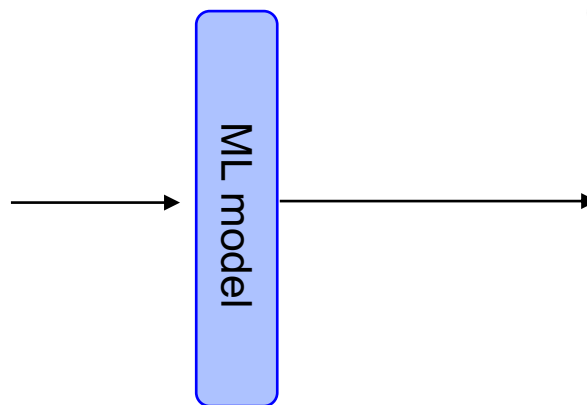
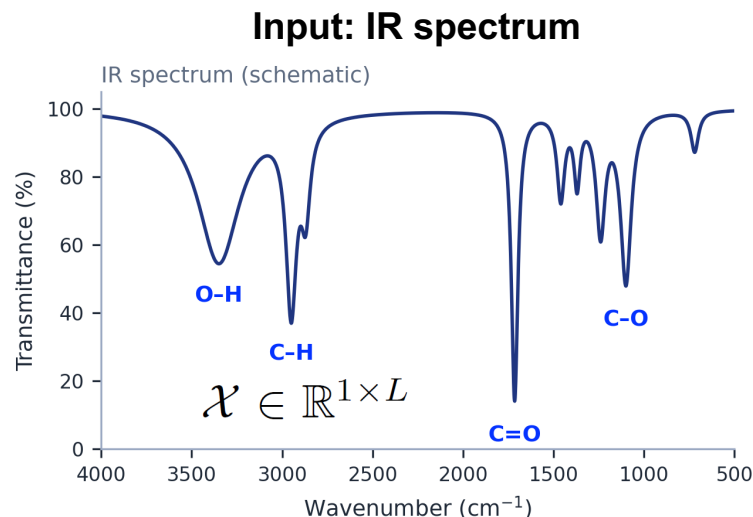
PCSP: the process that generates the spectrum

IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra

ICLR 2026

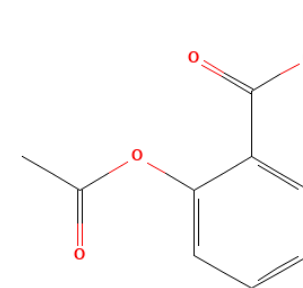
Heewoong Noh, Namkyeong Lee, Gyoung S. Na, Kibum Kim, Chanyoung Park

IR-Agent: from an IR spectrum to a full structure



Output: SMILES (Full molecular structure)

CC(=O)OC1=CC=CC=C1C(=O)O



▪ **Task: IR spectrum in, SMILES (the full molecular structure) out**

▪ **Existing approaches**

- Chemists read the peaks against an IR absorption table; it takes experience and intuition
- Most machine learning stops at functional-group detection: "there is a carbonyl", but not the molecule
- Methods that reach the full structure: Transformer sequence-to-sequence, spectrum in and SMILES out

IR-Agent: mirror how an expert reads a spectrum

- Experts read a spectrum as a combination, not peak by peak

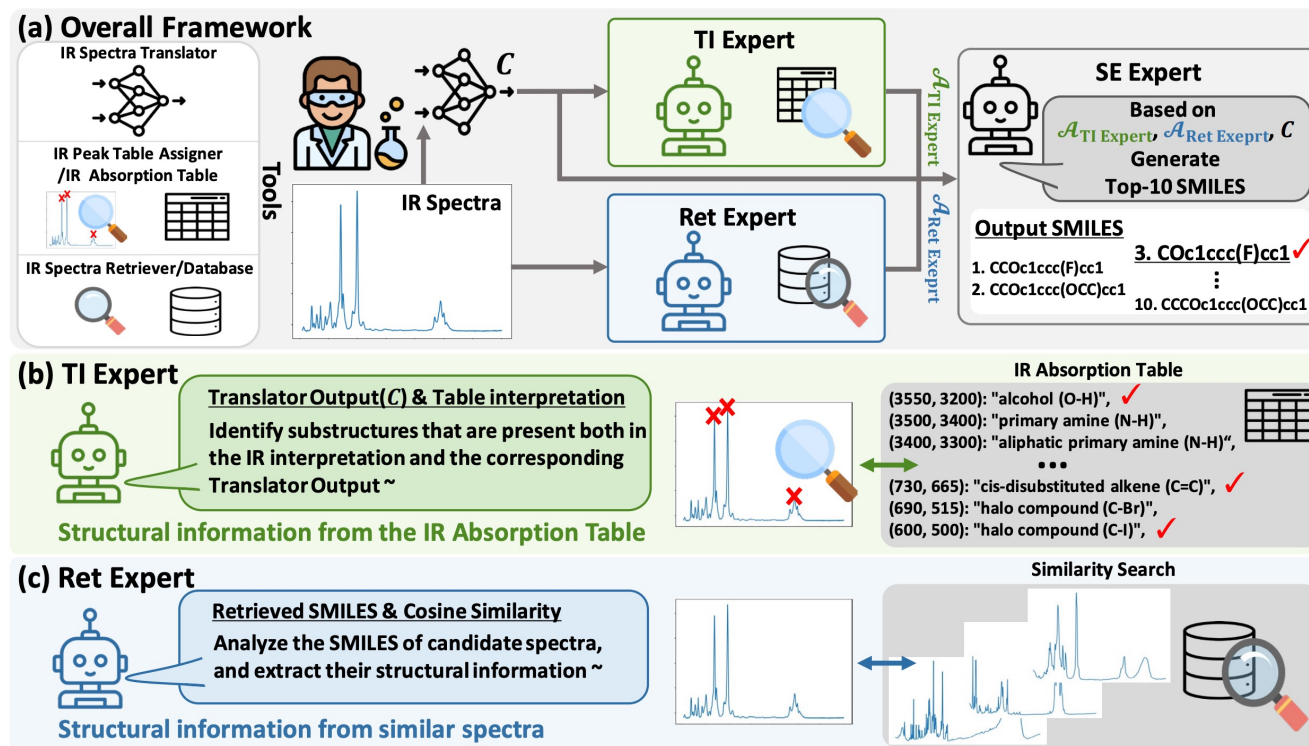
- Tools**

- Absorption table: peaks $\rightarrow$ substructures
- Retriever: nearest spectra $\rightarrow$ SMILES

- Three experts**

- TI expert (local): groups the peaks imply, with a confidence score
- Ret expert (global): structural motifs shared by similar spectra
- SE expert: integrates both and ranks the top-10 SMILES

- Translator proposes candidates first; the experts refine them



IR-Agent: results on 9k experimental NIST spectra

Method	Agent	Top-K Accuracy			
		Top-1	Top-3	Top-5	Top-10
Transformer	-	0.098 (0.007)	0.169 (0.000)	0.176 (0.003)	0.176 (0.003)
IR-Agent (GPT-4o-mini)	single	0.072 (0.008)	0.118 (0.002)	0.133 (0.002)	0.157 (0.003)
	multi	0.093 (0.005)	0.152 (0.003)	0.167 (0.005)	0.176 (0.005)
IR-Agent (GPT-4o)	single	0.083 (0.004)	0.135 (0.002)	0.165 (0.007)	0.194 (0.008)
	multi	0.093 (0.007)	0.153 (0.005)	0.177 (0.005)	0.204 (0.005)
IR-Agent (o3-mini)	single	0.087 (0.006)	0.153 (0.005)	0.179 (0.002)	0.197 (0.004)
	multi	0.103 (0.005)	0.178 (0.007)	0.199 (0.004)	0.216 (0.001)

*single: one LLM plays all the expert roles

Chemical Information	o3-mini				IR-Agent (single) (o3-mini)				IR-Agent (multi) (o3-mini)			
	Top-1	Top-3	Top-5	Top-10	Top-1	Top-3	Top-5	Top-10	Top-1	Top-3	Top-5	Top-10
No Knowledge	0.073 (0.010)	0.131 (0.011)	0.157 (0.011)	0.185 (0.005)	0.087 (0.010)	0.153 (0.011)	0.179 (0.011)	0.197 (0.005)	0.103 (0.005)	0.178 (0.007)	0.199 (0.004)	0.216 (0.001)
Scaffold	0.096 (0.002)	0.160 (0.006)	0.177 (0.003)	0.198 (0.003)	0.112 (0.003)	0.195 (0.009)	0.208 (0.008)	0.228 (0.010)	0.118 (0.003)	0.208 (0.009)	0.232 (0.008)	0.258 (0.010)
Carbon Count	0.105 (0.009)	0.158 (0.014)	0.186 (0.014)	0.214 (0.013)	0.121 (0.003)	0.177 (0.008)	0.194 (0.010)	0.219 (0.009)	0.123 (0.003)	0.190 (0.0005)	0.215 (0.009)	0.252 (0.007)
Atom Types	0.104 (0.011)	0.182 (0.007)	0.209 (0.005)	0.237 (0.003)	0.123 (0.006)	0.208 (0.003)	0.235 (0.011)	0.266 (0.009)	0.127 (0.006)	0.213 (0.003)	0.250 (0.011)	0.278 (0.009)

■ Structure prediction accuracy (Top-K)

- About 9k experimental IR spectra from NIST, three LLM backbones
- IR-Agent improves over the Transformer translator it starts from
- Multi-agent consistently beats a single agent playing all the roles

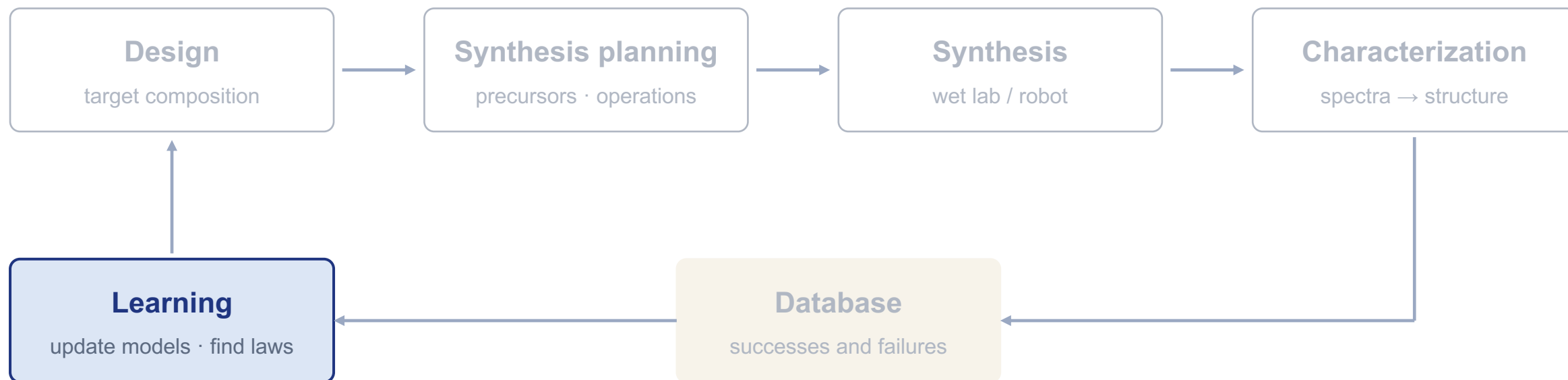
■ Adding chemical information

- Scaffold, carbon count, or atom types, added to the prompt as plain text
- No retraining, no architecture change, and accuracy goes up

→ The agent framework flexibly integrates chemical knowledge

Part 3 — Learning

▪ Part 3: **learning from the measurements** — CRISO



What should the loop learn from the data — model weights, or explicit laws?

Synthesis planning

- Retrieval-Retro (NeurIPS 2024)
- MSP-LLM (NeurIPS 2026)
- Retro-Forge (ICML 2026 AI4Sci WS)

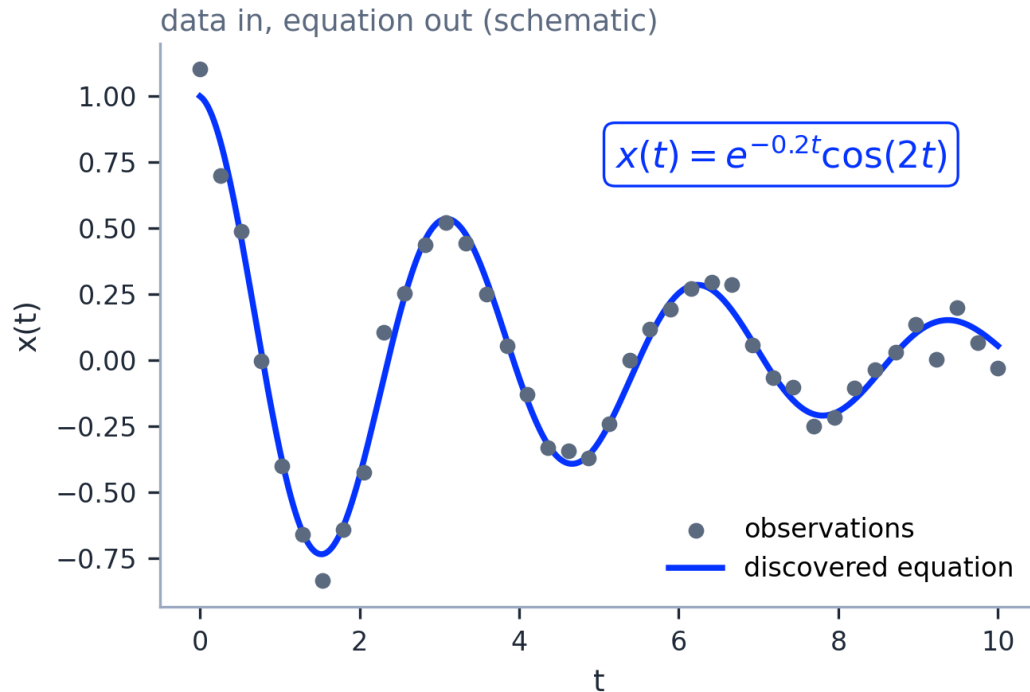
Characterization

- IR-Agent (ICLR 2026)
- PCSP (NeurIPS 2026)

Learning

- Multi-agent LLM equation discovery (under review)

Symbolic regression: from data to an equation



Observations in, an equation (structure + constants) out

- **Input: data. Output: a closed-form equation**

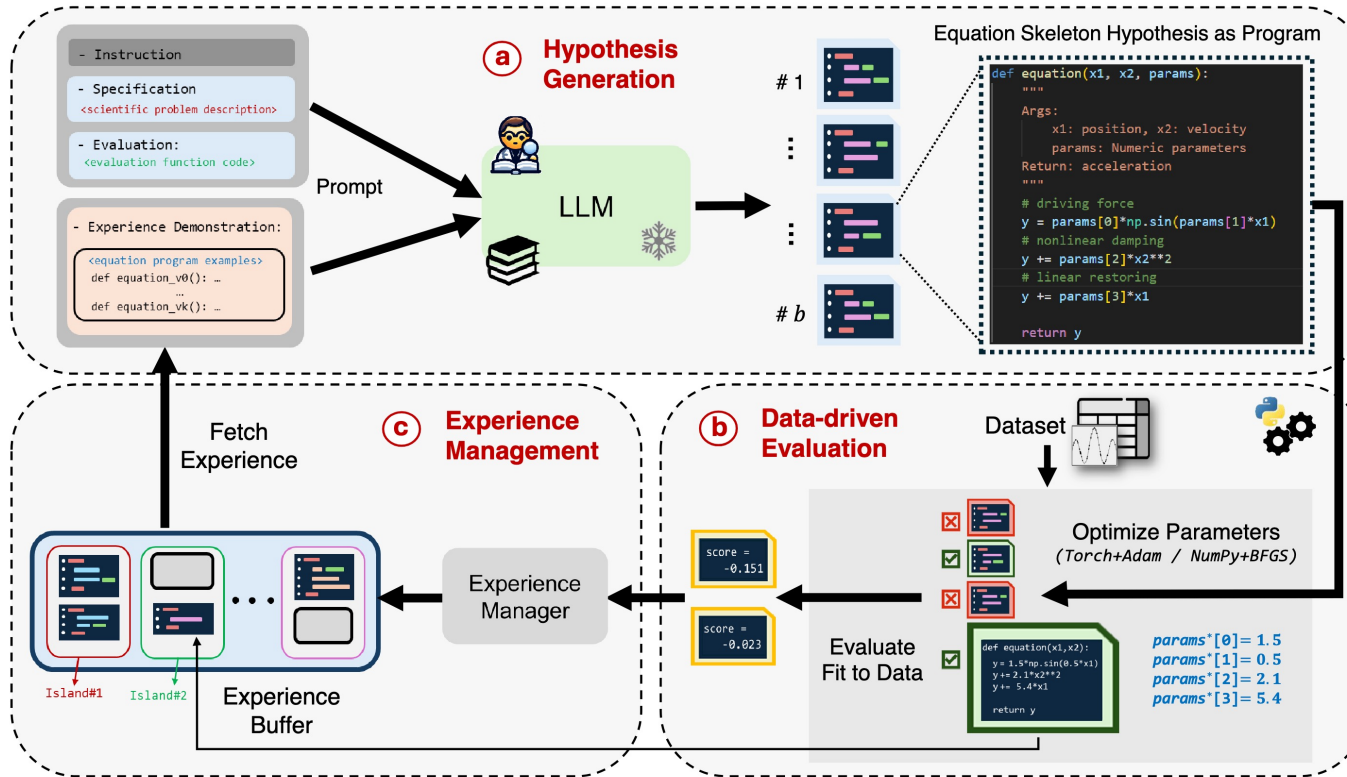
- Structure (operators, variables) plus constants
- Right structure: explains the system and extrapolates
- Right fit only: interpolates, and nothing more

- **Two families of methods**

- Classical SR (genetic programming, e.g. PySR): searches expression trees, but knows no physics
- LLM-based (e.g. LLM-SR): brings prior knowledge, but searches over text

→ **We want both: the LLM's knowledge and a structured symbolic search**

Background: LLM-SR (ICLR 2025 oral)



Step 1: Hypothesis Generation

```
def f(x, params):
    return params[0] * torch.sin(x) + params[1] * x**3
```

- Generates the equation skeleton only
- Good past equations are given in-context

Step 2: Data-driven Evaluation

$$\text{Score} = \text{MSE}(y - \hat{y})$$

- LLM proposes the structure; an optimizer (torch + Adam) fits the numbers

Step 3: Experience Buffer

- Good equations stored in a buffer (“equation:score”)
- Fed back as in-context examples
- LLM does mutation + cross-over

Three limits

- 1) Text-program search: no defined space, no symbolic operations
- 2) LLM prior: finds known equations, struggles on unseen systems
- 3) Error only: approximates the structure, poor extrapolation

CRISO: structured search by a collective of agents

- **1) Equations as abstract syntax trees (ASTs), not text**
 - A well-defined search space with symbolic operations
- **2) A collective of reasoning agents**
 - Many agents, each starting from a different hypothesis
 - Best equation $\rightarrow$ domain interpretation $\rightarrow$ shared with all agents $\rightarrow$ the collective learns structure outside any single agent's prior
- **3) Error and structure minimized together**
 - Score: prediction error + AST depth + number of free parameters
 - Prefers compact, explainable laws over fitted approximations
 - Finite-time convergence guarantee for the collective decision

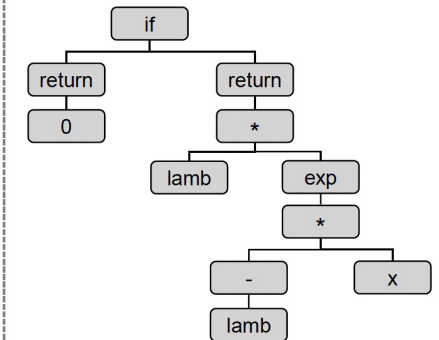
Informal description of scientific knowledge

PDF of an exponential distribution with rate $\lambda > 0$ is $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and otherwise 0. It starts at $f(0) = \lambda$ and decreases monotonically, capturing the idea that larger values ...

(a) Knowledge as a program

```
def equation(x, lamb):  
    # The distribution is defined only for non-  
    # negative values.  
    if x < 0:  
        return 0  
    else:  
        # 1. Calculate the exponent term  
        exp_term = -lamb * x  
  
        # 2. Compute the exponential function  
        exp_term = np.exp(exp_term)  
  
    return lamb * exp_term
```

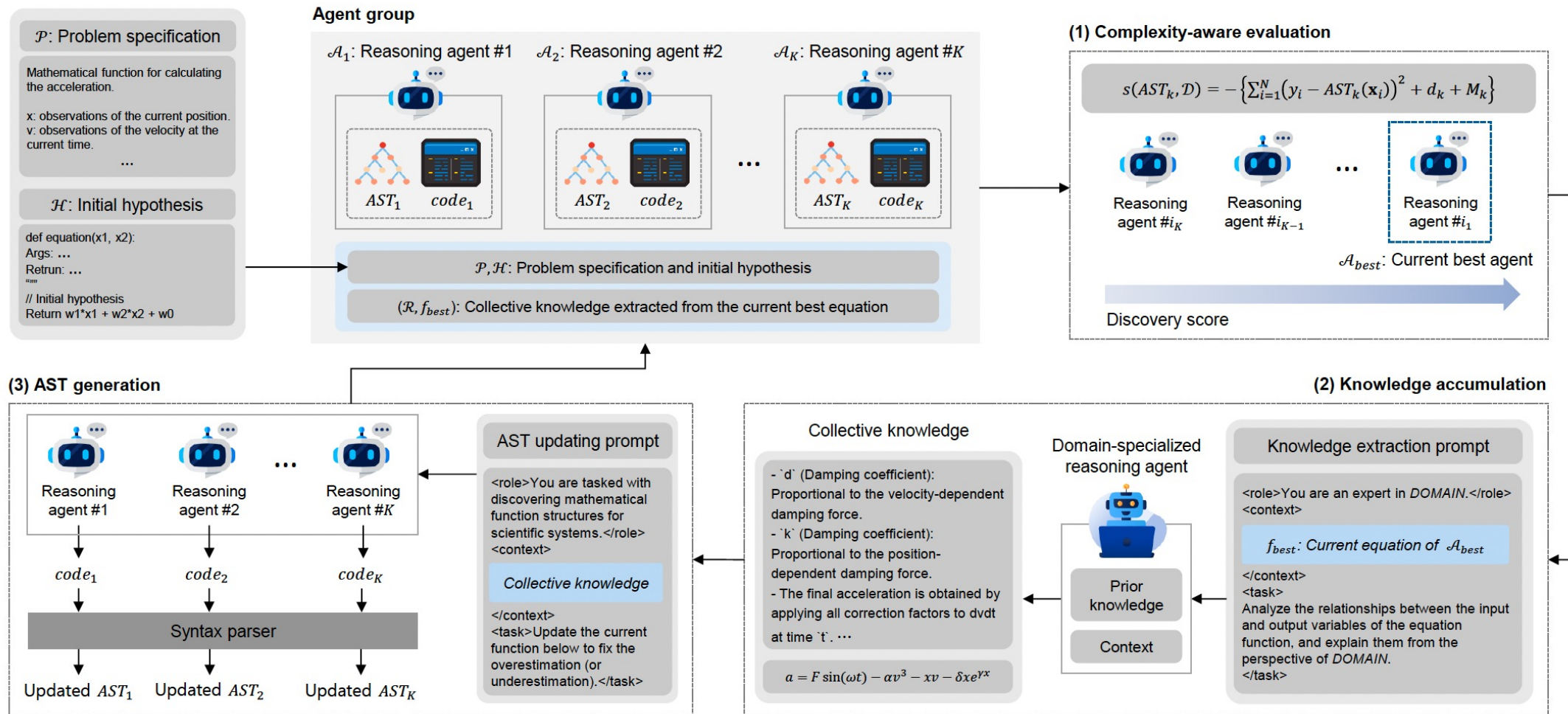
(b) Knowledge as an AST



$$s(AST_k, \mathcal{D}) = - \left\{ \sum_{(\mathbf{x}, y) \in \mathcal{D}} (y - AST_k(\mathbf{x}))^2 + d_k + M_k \right\}$$

d: AST depth, M: number of free parameters

CRISO: how the agent collective works



One round: (1) every agent's equation is scored on data and structure → (2) knowledge is extracted from the best one and shared with all agents → (3) each agent updates its AST

CRISO: largest gains where the equation is unknown

Domain	Problem Name (Target System)	Input Variables	Output Variable	Governing Equation
Physical science	Chi2PDF [34] (PDF of arbitrary chi-squared distributions)	x: Data point k: Degree of freedom	Probability density	$d_{x,k} = \frac{1}{2^{k/2}\Gamma(k/2)} x^{(k/2)-1} e^{-x/2}$
	NOSC [24] (Damped oscillator with nonlinear components)	t: Time x: Position v: Velocity	Acceleration	$a = 0.3 \sin(t) - 0.5v^3 - xv - 5xe^{0.5x}$
	NNN [1] (Nonlinear neural networks with sigmoid activation)	x_1: 1st model input x_2: 2nd model input	Model output	Fully-connected neural network with two hidden layers in Eq. (6)
	MSB [35] (Strain-stress relation in a metallic material)	T: Temperature ϵ: Strain	Mechanical stress	Unknown
Chemical science	FHST [36] (Flory–Huggins equation for polymer mixing)	ϕ: Volume fraction N: Chain length χ: Interaction param.	Total change of Gibbs free energy	$\Delta G_m = RT \left[\frac{\phi}{N} \ln \phi + \psi \ln \psi + \chi \phi \psi \right]$, where $R = 8.314, T = 300, \psi = 1 - \phi$
	BDC [37] (Lithium-ion battery)	Six measurement parameters (cycle, voltage, etc.)	Discharge capacity	Unknown
	SFL [37] (Fatigue limit of alloy compositions)	Weight Percents of ten elements and tempering temperature	Fatigue limit	Unknown
	NOMC [38] (Chemical reactor for non-oxidative methane conversion)	Six engineering parameters (pressure, time, etc.)	C ₂ conversion yield	Unknown
	ECBG [24] (Differential equation for bacterial growth rate of Escherichia coli)	b: Population density s: Substrate conc. T: Temperature pH: pH level	Bacterial growth rate	Differential equation for bacterial growth rate in Eq. (9)
Biology	HHM [39] (Differential equation for membrane potential in Hodgkin–Huxley model)	Five biophysical parameters (membrane potential, gating probability, etc.)	Rate of potential change	Differential equation for Hodgkin–Huxley membrane potential dynamics in Eq. (??)

Problem	PySR	GenSR	LLM-SR	FuncEv	STRIDE	CRISO
Chi2PDF	0.4587 (±0.0199)	0.1752 (±0.0113)	0.0289 (±0.0013)	0.0897 (±0.0061)	1.05E-5 (±4.48E-6)	5.67E-6 (±1.87E-6)
FHST	0.0783 (±0.0091)	0.0403 (±0.0012)	0.0147 (±0.0010)	0.0117 (±0.0008)	0.0375 (±0.0010)	9.40E-8 (±1.62E-8)
HHM	0.8869 (±0.0129)	0.1062 (±0.0069)	1.85E-6 (±2.13E-6)	0.0834 (±0.0051)	3.51E-7 (±1.26E-7)	7.93E-8 (±4.79E-8)
NDO	0.1713 (±0.1741)	0.0661 (±0.0036)	0.0289 (±0.0014)	0.0010 (±0.0002)	N/A	6.92E-5 (±0.0001)
NNN	N/A	0.0650 (±0.0041)	0.0050 (±0.0002)	0.0059 (±0.0003)	0.0030 (±0.0002)	0.0016 (±0.0001)
ECBG	1.4973 (±0.0001)	0.8828 (±0.0326)	0.3315 (±0.0133)	0.9499 (±0.0034)	0.4979 (±0.0338)	0.8742 (±0.0073)
MSB	0.1613 (±0.0101)	0.0449 (±0.0058)	0.0418 (±0.0018)	0.0402 (±0.0013)	0.0887 (±0.0049)	0.0299 (±0.0028)
BDC	0.0149 (±0.0035)	0.0143 (±0.0023)	0.0128 (±0.0003)	0.0098 (±0.0004)	0.0124 (±0.0003)	0.0092 (±0.0011)
SFL	0.1179 (±0.0016)	0.0585 (±0.0045)	0.1078 (±0.0050)	0.0545 (±0.0022)	0.1925 (±0.0046)	0.0431 (±0.0011)
NOMC	0.2031 (±0.0001)	0.1269 (±0.0158)	0.1409 (±0.0081)	0.1252 (±0.0135)	0.1735 (±0.0156)	0.0776 (±0.0154)

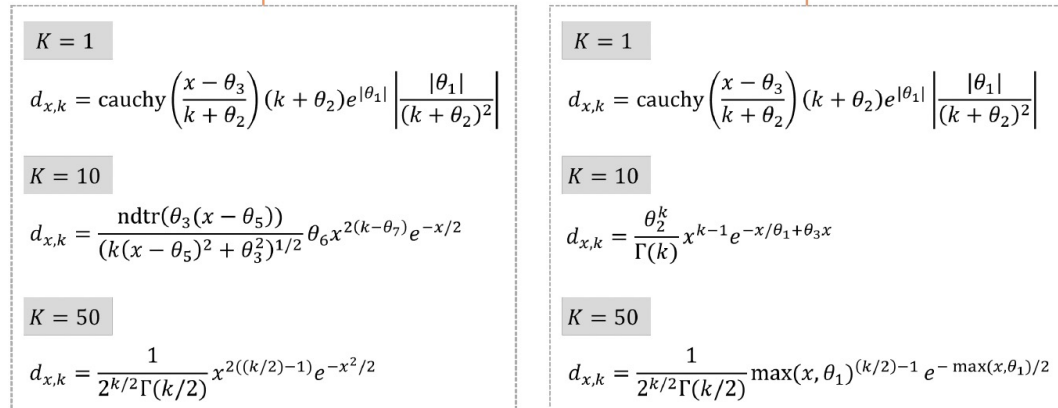
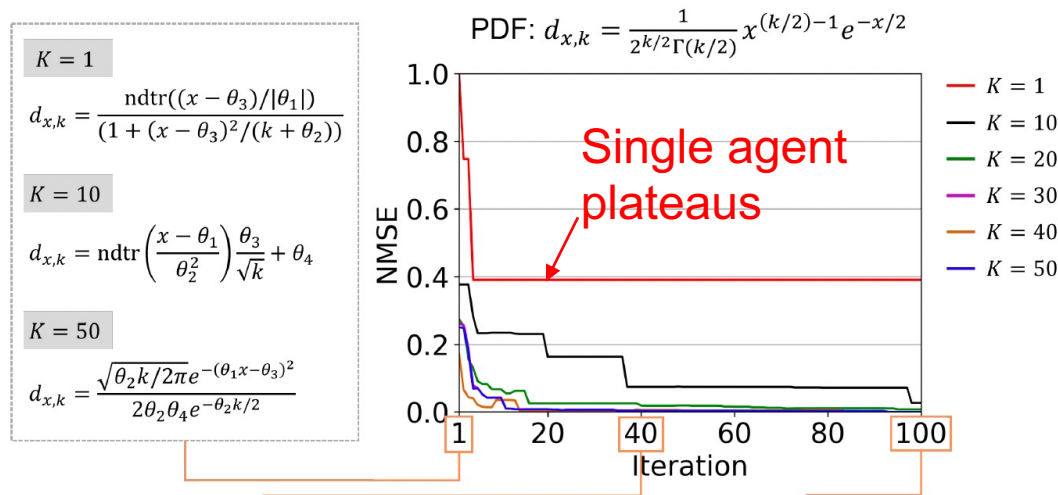
Error, lower is better

- Consistent gains over LLM-SR and classical SR
- Largest gains where the equation is unknown
- Recovered equations are compact and readable

10 systems (physics, chemistry, biology); 4 with unknown equations

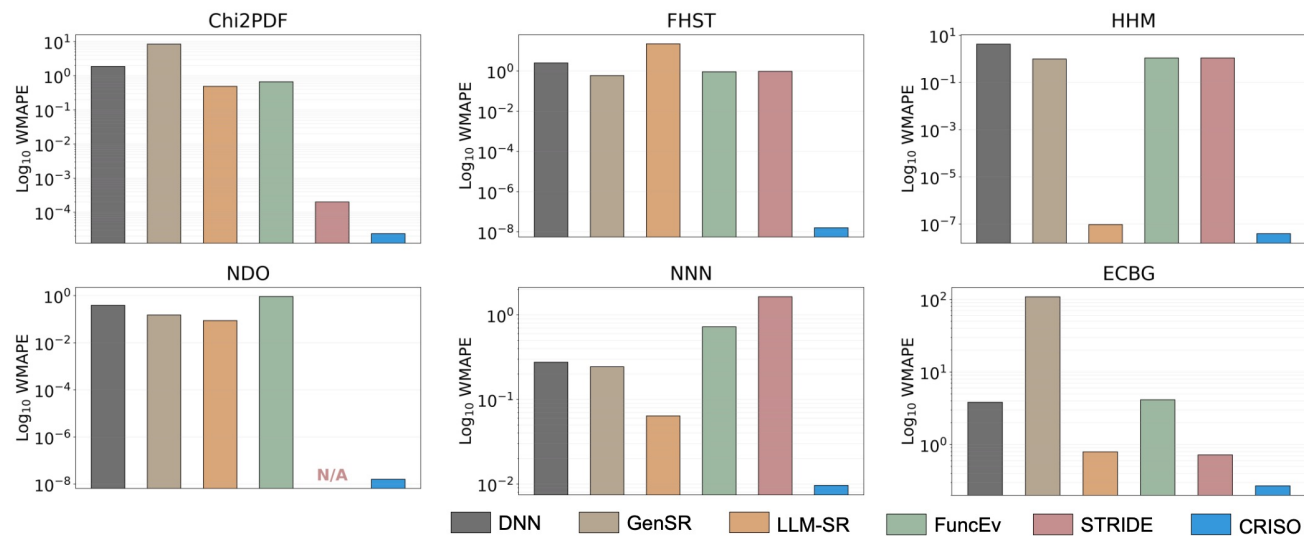
CRISO: discovered laws hold out of distribution

Progressive update



As K grows, the true equation's structure is recovered

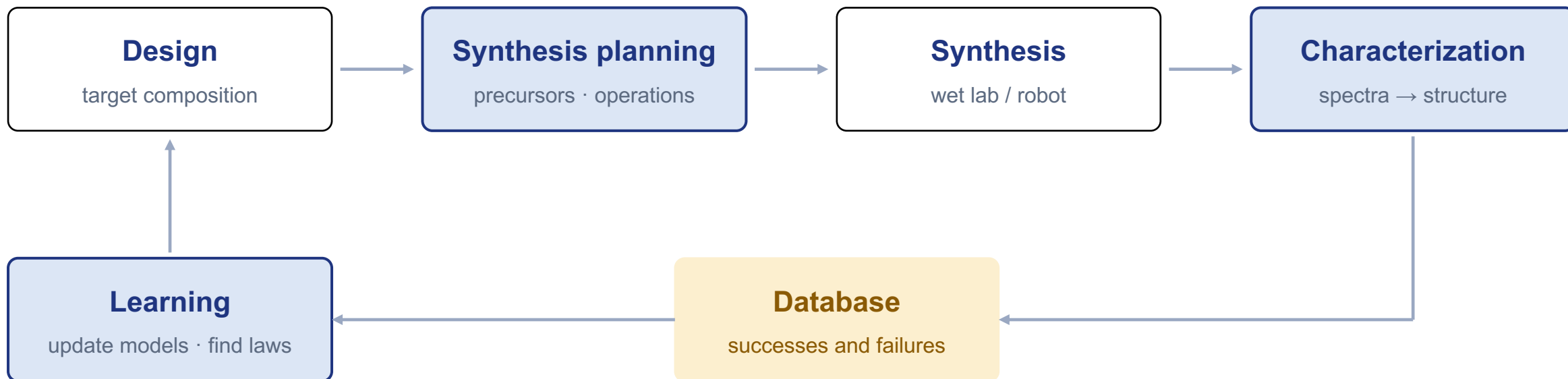
Out-of-distribution



- DNNs fail badly on some problems, especially OOD
- LLM-SR also shows a large error increase OOD
- existing methods fail at extrapolation
- In contrast, CRISO has much lower error, especially OOD
- strong OOD generalization from structural law discovery

Summary: AI across the materials discovery loop

- Three stages of the loop, **three AI systems**, one common thread



Synthesis planning

Retrieval-Retro · MSP-LLM · Retro-Forge

Target → recipe: retrieve related recipes, plan with an LLM, decompose step by step

Characterization

IR-Agent

Spectrum → structure: read it like an expert, local and global cues combined

Learning

CRISO

Data → law: a collective of agents searches for the equation, not only a fit

References

- (NeurIPS 2024) Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge
- (NeurIPS 2026) A Structured LLM Framework for Inorganic Material Synthesis Planning
- (ICML 2026 AI for Science Workshop) Retro-Forge: A Multi-Step Pairwise Retrosynthesis Framework for Solid-State Materials Synthesis
- (ICLR 2026) IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra
- (NeurIPS 2026) Periodic Complex Stochastic Processes for Retrieving Atomic Structures of Unknown Matters
- (Under review) CRISO: Machine Collective Intelligence for Scientific Discovery

Thank you

cy.park@kaist.ac.kr · dsail.kaist.ac.kr

